$ite : \{A : Ty\} \to Tm\ A \to (Tm\ A \to Tm\ A) \to Tm\ Nat \to Tm\ A$

$$\underbrace{ite\ \{Nat \Rightarrow Nat\}\ \underline{(lam\ \lambda x.x)}\ (\lambda f.\ \underbrace{lam\ \lambda n.f\ n}_{=\lambda f.f})\ \underline{(suc\ (suc\ zero))}}_{: Tm\ (Nat \Rightarrow Nat)} \cdot suc\ zero$$

$lam\ (\lambda x.x) \cdot suc\ zero$
$= suc\ zero$

$ite\ z\ s\ zero = z$
$ite\ z\ s\ (suc\ n) = s\ (ite\ z\ s\ n)$
$ite\ z\ s\ (suc\ (suc\ zero)) = s\ (ite\ z\ s\ (suc\ zero)) = ite\ z\ s\ (suc\ zero) =$
$ite\ z\ s\ zero = lam\ \lambda x.x$

0	$f \mapsto \lambda x.x$
1	$f \mapsto f$
2	$f \mapsto f \circ f$
3	$f \mapsto f \circ f \circ f$
.	
n	$f \mapsto f^n$

$ite\ \{(A \Rightarrow A) \Rightarrow (A \Rightarrow A)\}$
$(lam\ \lambda f.\ lam\ \lambda x.x)$ z
$(\lambda h.\ lam\ \lambda f.\ lam\ \lambda x.\ f \cdot (h \cdot f \cdot x))$ s

$ite\ z\ s\ zero = lam\ \lambda f.\ lam\ \lambda x.\ x$
$ite\ z\ s\ (suc\ zero) =$
$lam\ \lambda f.\ lam\ \lambda x.$
$f \cdot ((lam\ \lambda f.\ lam\ \lambda x.x) \cdot f \cdot x)$
$\| \Rightarrow \beta, 2x$
$lam\ \lambda f.\ lam\ \lambda x.$
$f \cdot (x)$

System T (ite)-ben nem lehet megadni
$pred : Tm\ (Nat \Rightarrow Nat)$, melyre $pred \cdot (suc\ x) = x$ $\quad x : Tm\ Nat$

System T (ite)-ben a $\underbrace{-1}_{: \mathbb{N} \to \mathbb{N}}$ függvény definiálható?
$pred : Tm\ (Nat \Rightarrow Nat)$, melyre $\forall n : \mathbb{N}$-re $pred\ \ulcorner 1+n \urcorner = \ulcorner n \urcorner$
$\|$
$pred\ (suc\ \ulcorner n \urcorner)$

$(lam\ \lambda x.\ \boxed{x})\ Tm\ (Nat \Rightarrow Nat)$
$\uparrow$ $Tm\ Nat$

$\boxed{lam\ \lambda x.\ pred \cdot (suc\ x)} \neq lam\ \lambda x.\ x$

System T rekurzorral
System T − ite + rec

$rec : Tm\ A \to (\underline{Tm\ Nat} \to Tm\ A \to Tm\ A) \to Tm\ Nat \to Tm\ A$
$rec\ z\ s\ zero = z$
$rec\ z\ s\ (suc\ n) = s\ n\ (rec\ z\ s\ n)$

$pred : Tm\ Nat \to Tm\ Nat$
$pred := rec\ zero\ (\lambda n\ x.\ n)$
$pred\ (suc\ x) = rec\ zero\ (\lambda n\ x.n)\ (suc\ x) =$
$(\lambda n\ x.n)\ x\ (rec\ \_\ \_\ x) = x$

Van olyan eval $\mathbb{N} \to \mathbb{N} \to \mathbb{N}$ függvény, amely System T-ben nem definiálható.

$code : Tm\ (Nat \Rightarrow Nat) \to \mathbb{N}$
$\ulcorner eval\ (code\ t)\ n \urcorner = \underbrace{t \cdot \ulcorner n \urcorner}_{: Tm\ Nat}$

$V : Tm\ Nat$, melyre $V = suc\ V$.
$u : Tm\ (Nat \Rightarrow Nat)$
$u := lam\ \lambda x.\ suc\ (t \cdot x \cdot x)$

eval def-ható: $\exists t.$
$\ulcorner eval\ m\ n \urcorner = t \cdot \ulcorner m \urcorner \cdot \ulcorner n \urcorner$
$\to t \cdot \ulcorner code\ u \urcorner \cdot \ulcorner code\ u \urcorner =$ (def-ható)
$\ulcorner eval\ (code\ u)\ (code\ u) \urcorner =$ (eval kiértékelő)
$u \cdot \ulcorner code\ u \urcorner =$ (u def.)
$(lam\ \lambda x.\ suc\ (t \cdot x \cdot x)) \cdot \ulcorner code\ u \urcorner \overset{\Rightarrow\beta}{=}$
$suc\ (t \cdot \ulcorner code\ u \urcorner \cdot \ulcorner code\ u \urcorner)$

$|\mathbb{N} \to 2| > |\mathbb{N}|$ $\boxed{xkcd}$ fetishes
$(\mathbb{R} > \mathbb{N})$ Andrej Bauer Countable reals

0. 0 0 0 0 . . . .
1. 0 1 0 1 0 . . . .
2. 1 1 1 1 1 1 1 0 0 0 . .
3. 1 0 0 0 0 0 0 0 0 0
.
.
$\boxed{1\ 0\ 0\ 1\ \ldots}$ ← nincs benne

$\underbrace{\_\Rightarrow\_,\ Nat,\ \_\times\_}_{: Ty \to Ty \to Ty}$ $\quad$ Szorzat, rekord, osztály

$Tm\ A \times Tm\ B \cong Tm\ (A \times B)$
$\_,\_ : Tm\ A \to Tm\ B \to Tm\ (A \times B)$
$fst : Tm\ (A \times B) \to Tm\ A$
$snd : Tm\ (A \times B) \to Tm\ B$
$\times\beta_1$: $fst\ (a,b) = a$
$\times\beta_2$: $snd\ (a,b) = b$
$\times\eta$: $w = (fst\ w,\ snd\ w)$

obj	mat
Nat	$\mathbb{N}$
$\Rightarrow$	$\to$
Bool	2
zero	0
suc zero	1
$\_\times\_$	$\_\times\_$

$Nat \times Nat \Rightarrow Nat$
$Nat \Rightarrow (Nat \Rightarrow Nat)$
$Nat \Rightarrow Nat \times Nat$
$Nat \Rightarrow Nat$
$Nat \Rightarrow Nat$

$range : Tm\ (Nat \Rightarrow Nat \times Nat)$
$range := lam\ \lambda x.\ (pred \cdot x,\ suc\ x)$

Haskell Curry ötlete

$(a,b) \mapsto (b,\ suc\ b)$

pred:		
0	$\boxed{0}$	(0,0)
1	0	(0,1)
2	1	(1,2)
3	2	(2,3)
4 ...	3	(3,4)

$fst\ (ite\ \{Nat \times Nat\}\ (zero,\ zero)\ (\lambda w.\ (snd\ w,\ suc\ (snd\ w)))\ n)$

$pred\ (suc\ \ulcorner 3 \urcorner)$
$\|$
$\ulcorner 3 \urcorner$
$pred\ (suc\ x)$
$\neq$
x